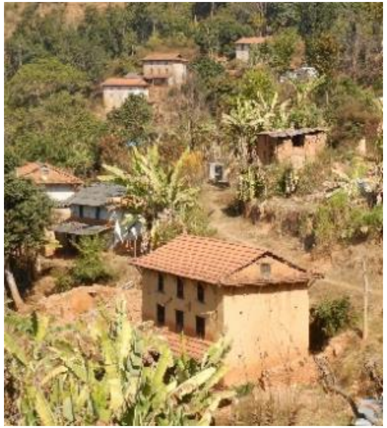
# SEISMIC RETROFITTING OF UNSAFE HOUSING IN NEPAL

2017

# Common Buildings in Affected Districts

**Makwanpur**

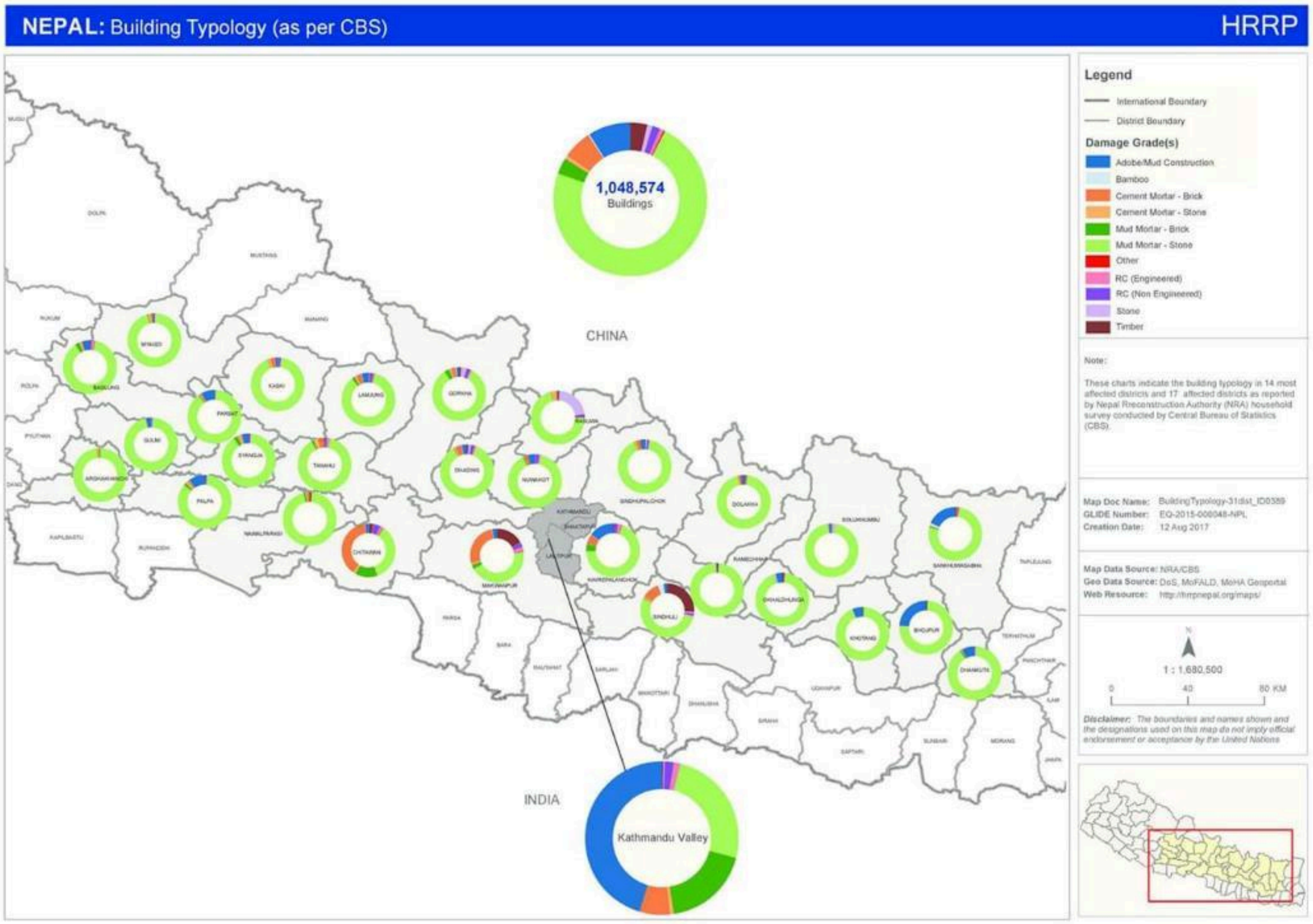
**Sindhuli**



**Sindhupalchok**

**Kavre**

# Building Typologies in 31 Districts



# Seismic Retrofit

- Seismic Retrofit
  - The modification of existing structures to make them more resistant to seismic activity or ground motion due to earthquakes.

# Government Approved Douments for Retrofitting

Retrofit Type design on stone in mud mortar building – Light weight

# Retrofit Type design on stone in mud mortar building – heavy weight

Retrofit Type design on stone in mud mortar building – Heavy weight

# Retrofit Design Principles

- Use locally available resources whenever possible
- Preserve local architecture and construction heritage where possible
- Minimize impacts to functionality of the interior space
- Optimize costs by utilizing the strength of existing structure
- Consideration of prevalent hazards other than earthquakes

# Retrofit Design Principles

- Avoid complexity
  - Introduce as few changes as possible to achieve goal
    - Mobilize strength of existing structure
  - Avoid relying on flexible roof diaphragm systems
  - Don't depend on things that can't be easily checked (through stones)
- Consider other hazards / deficiencies where possible
  - Anchor roofs, verandahs, canopies, etc.

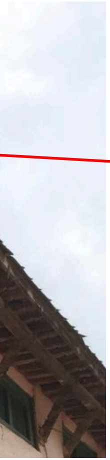
Follow the three C's

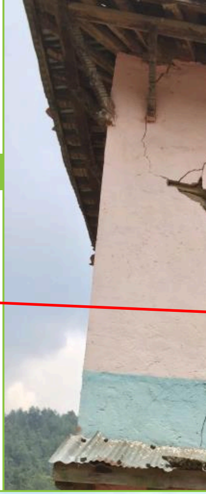
- Configuration
  - Regularity
- Connections
  - "Tie the building together"
  - Shift behavior from non-ductile to ductile lateral deformation modes
- Construction Quality
  - Use good quality materials

# Typical SMM Buildings

# Typical Observed Deficiencies

# Typical Observed Deficiencies

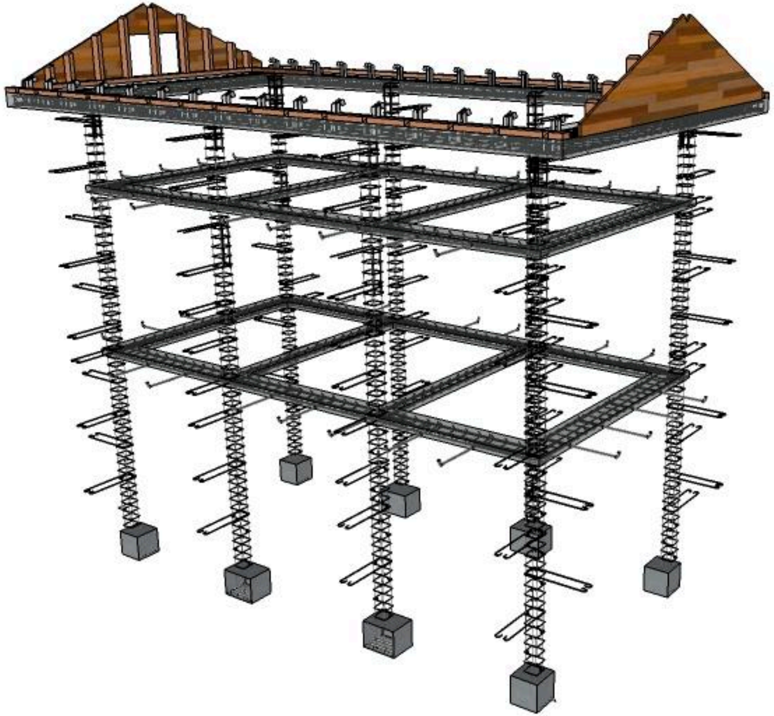




1. Wall Delamination
2. Gable End Wall Damage
3. Short Walls Pull Away from Horizontal Diaphragms
4. Parapets and Cantilever Walls on Long Sides
5. Horizontal Diaphragms Deform in Bending / Shear

# Retrofitting Elements

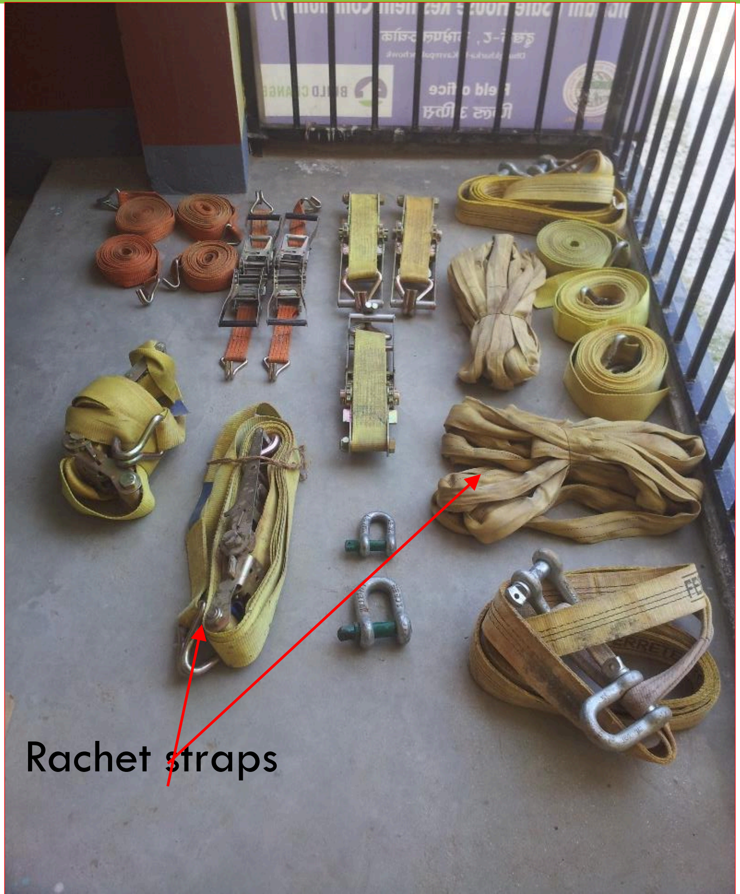
# Retrofit elements



# Retrofitting Process

# Shoring

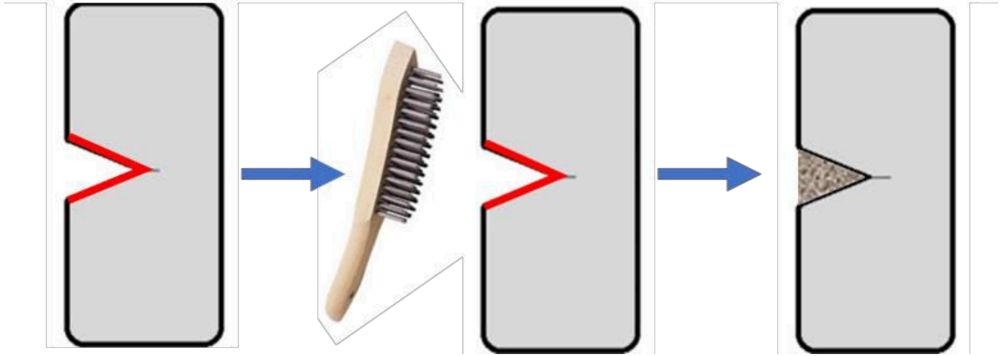
- **Shoring** is the process of supporting a building, vessel, structure, or trench with shores (props) when in danger of collapse or during repairs or alterations. *Shoring* comes from *shore* a timber or metal prop.



# Shoring

# Repair of cracks

- **Cleaning of crack and grouting with stabilized mud mortar (1:4:5 – Cement:sand:mud)**

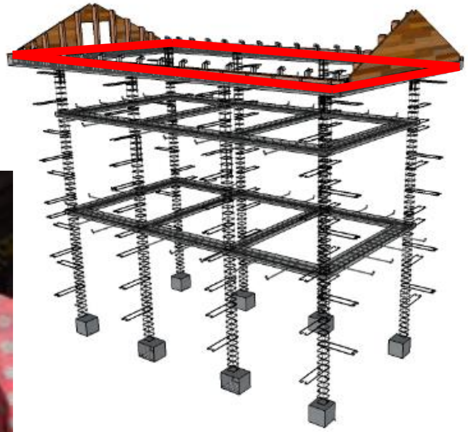


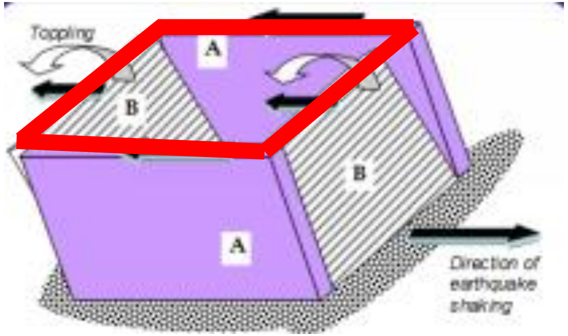
Making ‘V’ Notch

Cleaning crack with wire brush

Sealing crack with Stabilized mortar (1:4:5)

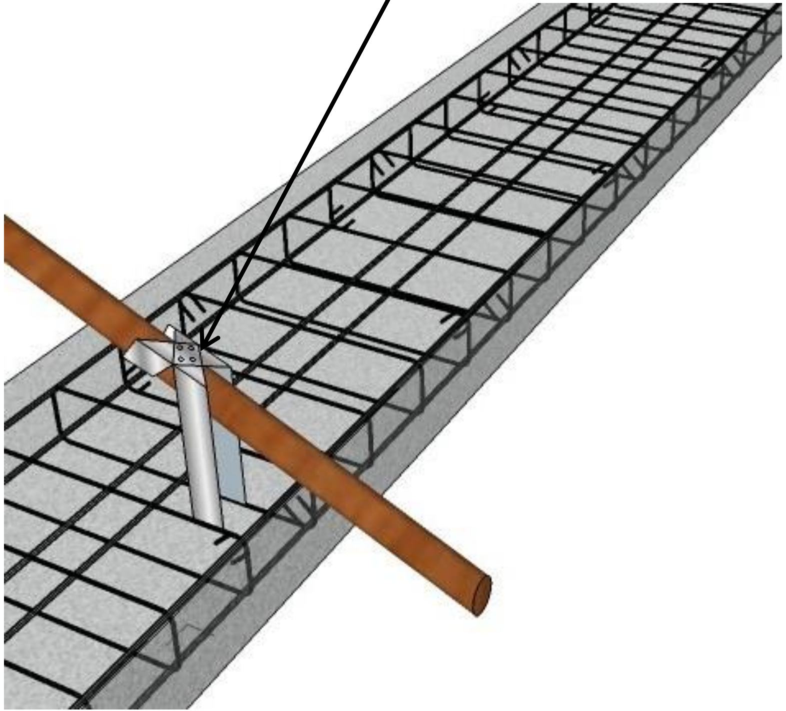
# Ring Beam





# Connection of Roof to Ring Beam

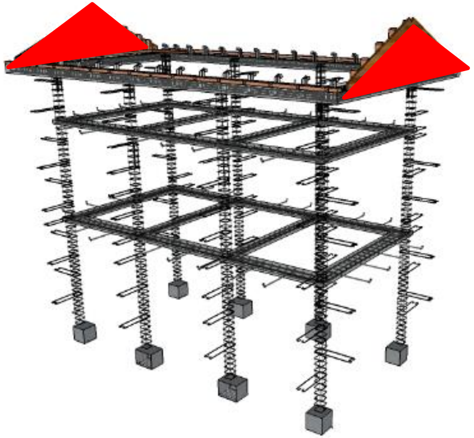
Strap embedded in concrete to bind rafter



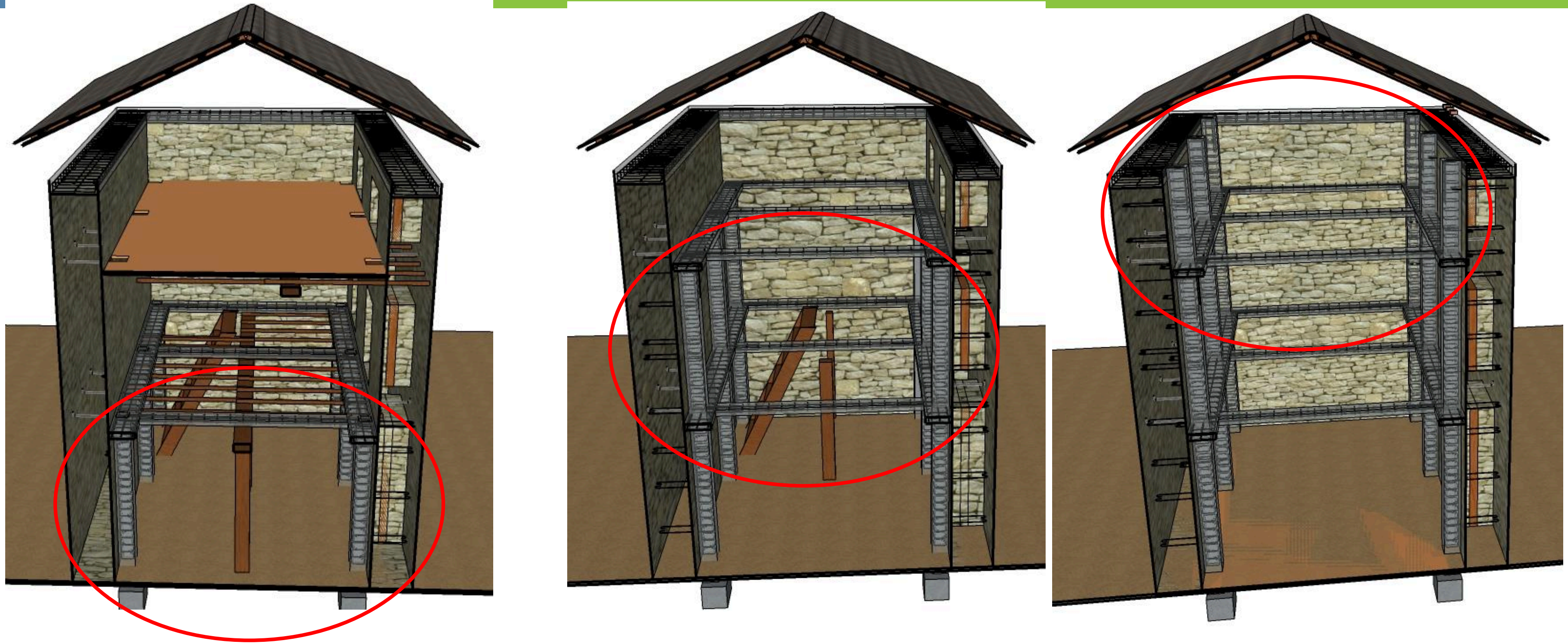
*Roof Detail*

# Connection Improvement: Roof

# Light Gable

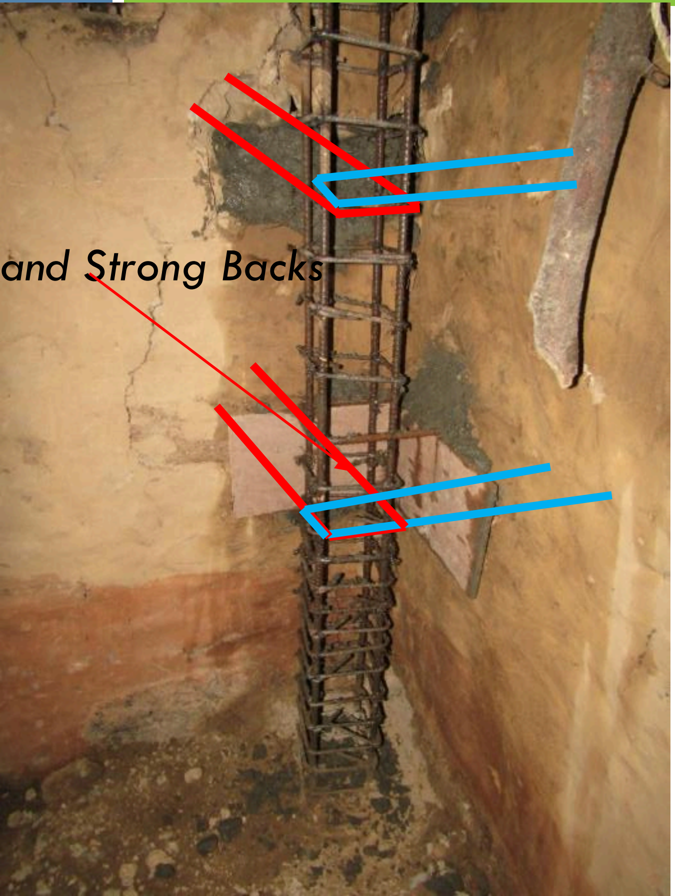


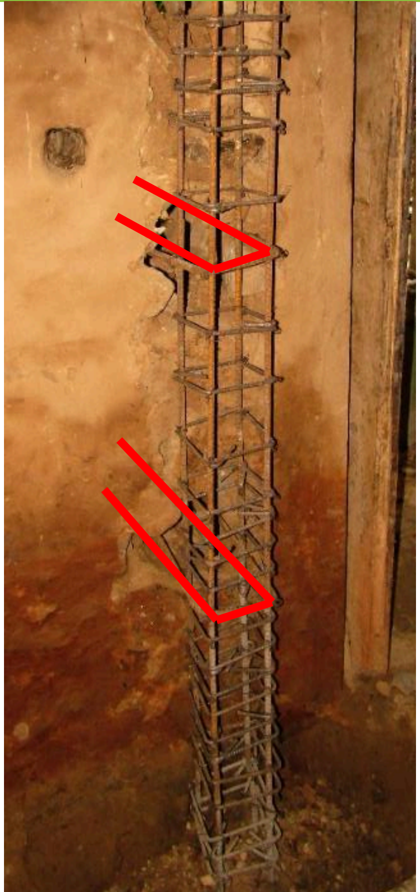
# Strong backs placement in sequence for heavy weight scheme

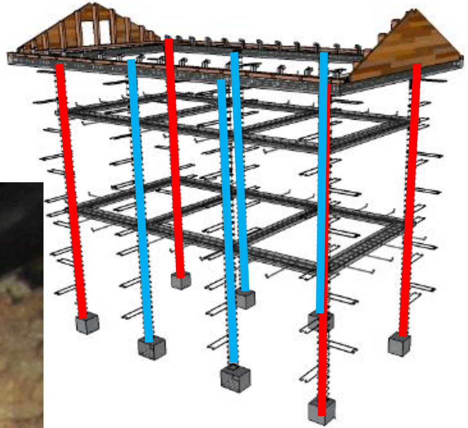


*Strong backs are provided at the corners and at the middle of wall piers to improve connectivity and brace the walls against out of plane forces*

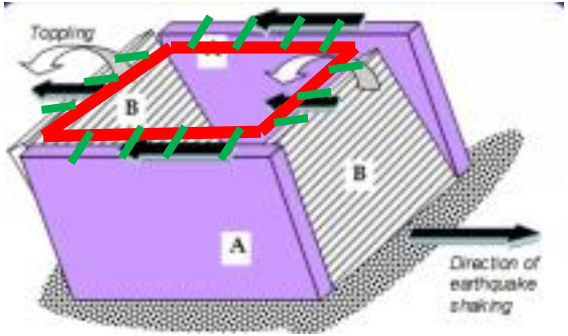
# Strong Backs: Reinforced Concrete

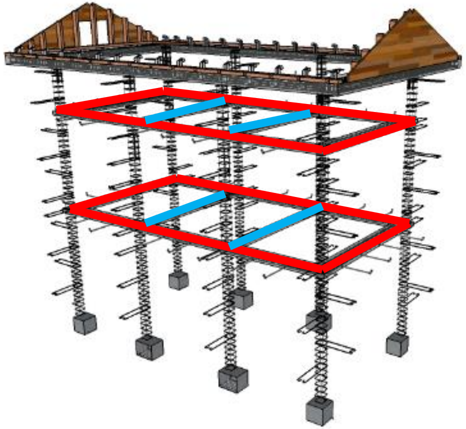




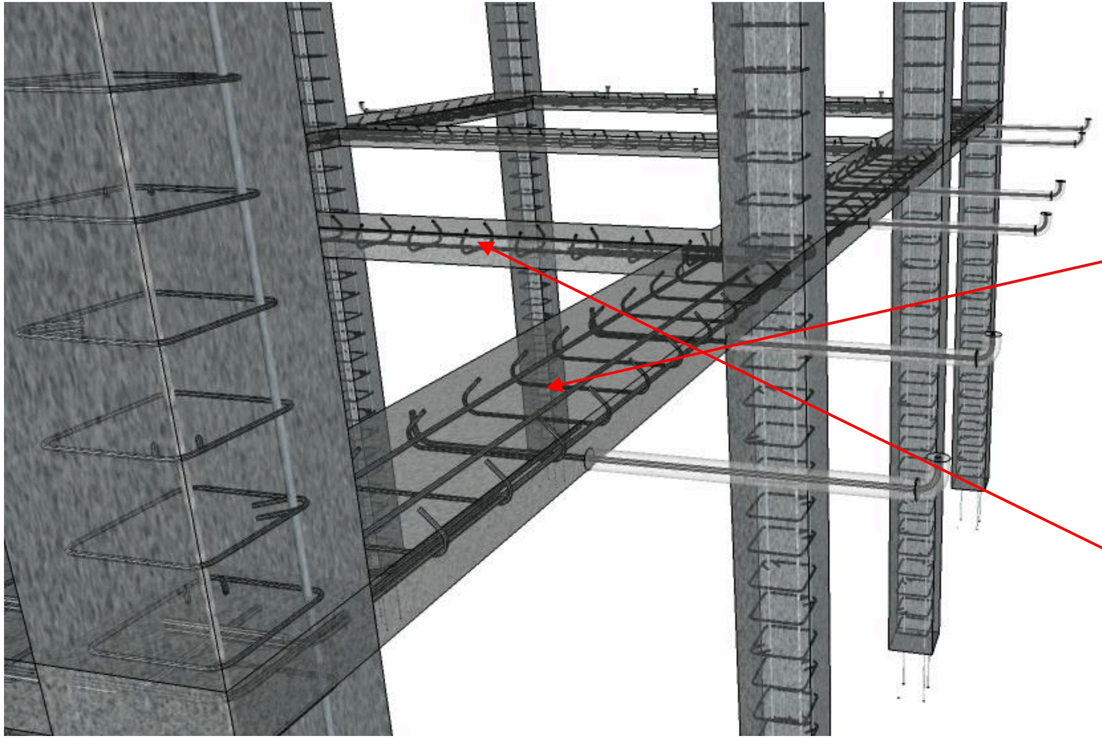
# Slab Strip



*e provided around the perimeter of the walls on the inside and across connecting opposite strong backs*
*wall*

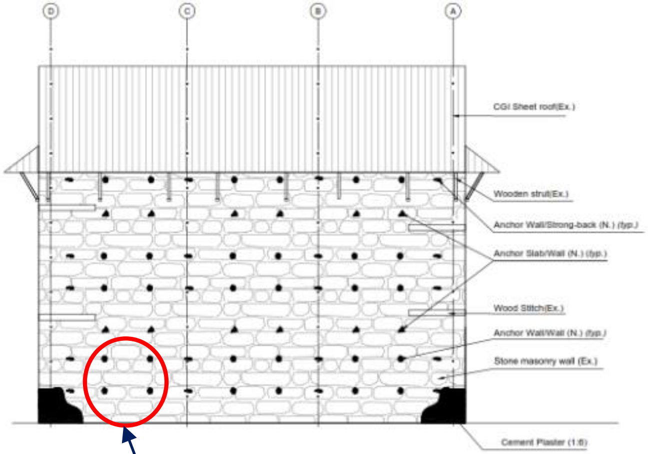
# Strengthening of the diaphragm for heavy weight scheme

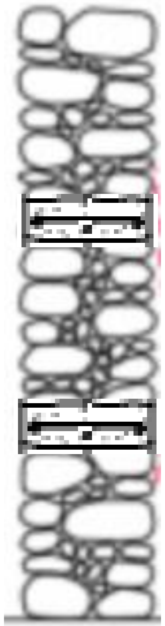
*Slab strips are provided around the perimeter (300mm) of the walls on the inside and across connecting opposite strong backs (200mm) along the longitudinal wall*

*3-12mm longitudinal rebars in perimeter and 2-12mm longitudinal rebars in mid strip and 7mm hooks spaced at 150mm center to center in perimeter*

# Strengthening of Wall: Through Concrete




Through wall concrete



# Strengthening of Wall: Plaster

# Retrofitted Building

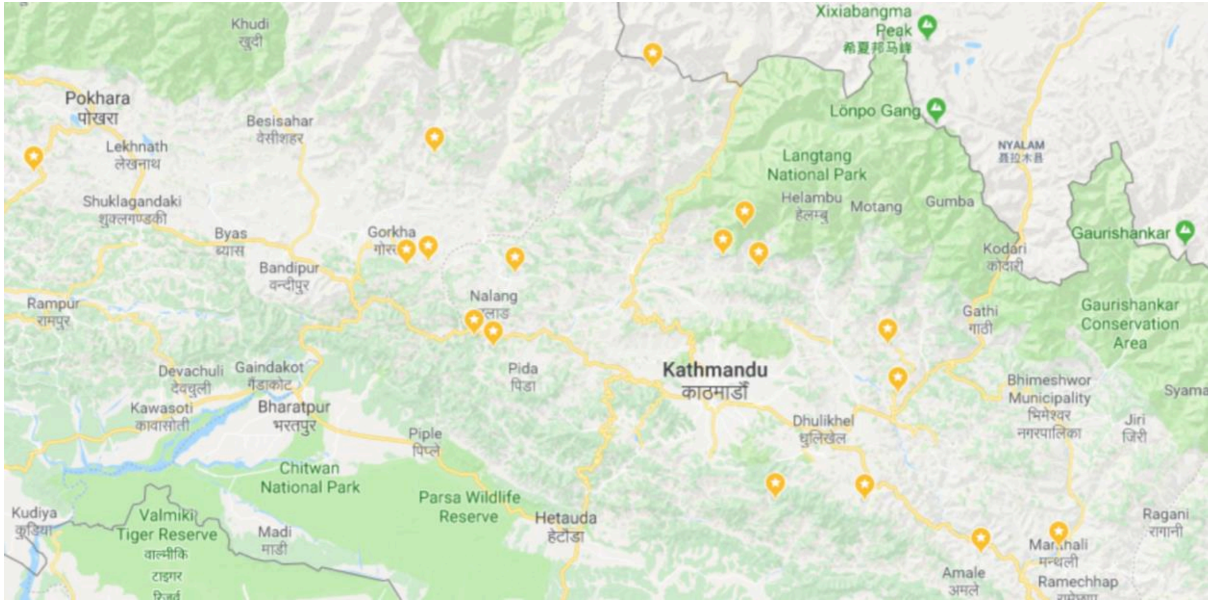
## House before the retrofit

## House after the retrofit

# SPRINGBOARD PROJECT

# Springboard Project – Geographic Spread

- 36 retrofitted houses, 17 locations, 9 districts
- Mid-hills & northern areas of the 14 most earthquake districts
- Additional 17 districts




*Map of 17 locations*

# Springboard Project – Example Retrofit

Retrofitted house at Shaurpani, Gorkha in partnership with Oxfam and local partners

# Springboard Project – Take-Up Models

- Take-up models;
  - Home owner driven
  - Entrepreneur and private sector driven
  - Partner organization driven
  - Type design development
  - Training development

*A trainee builder practicing shoring as a part of OJT in Chyamurunbesi, Kavre delivered in partnership with Helvetas Employment Fund and private sector service provider NETSA*

# Springboard Project – GoN Approval

- Base type design for stone with mud mortar (SMM) #1 heavyweight and lightweight options successfully approved by NRA Technical Committee and MoUD CLPIU
  - Including go/no-go assessment and tranche release inspection checklist etc.
- Further type designs developed, piloted and submitted to CLPIU for approval including;
  - Expanded customisation options for SMM
  - Dry stone heavyweight and lightweight options

# Springboard Project – Training

- ≈100 engineers participated in 7-day training on the initial SMM type design or received on site coaching
- ≈2,000 NRA/DUDBC field inspectors received orientation training on the initial type design and related tools (go/no-go assessment and tranche release inspection checklist etc.)
- ≈Additional 1,000 NRA/DUDBC field inspectors received the orientation training through adjacent funding
- ≈200 local builders participated in on-the-job training to deliver the 36 retrofits
- Finalization of mason training curricula is currently ongoing together with Helvetas Employment Fund.

# Springboard Project – Training

Builder Trainer delivering retrofitting on-site coaching to a trainee builder in Sangachok, Sindhupalchok with partner organisation WVIN

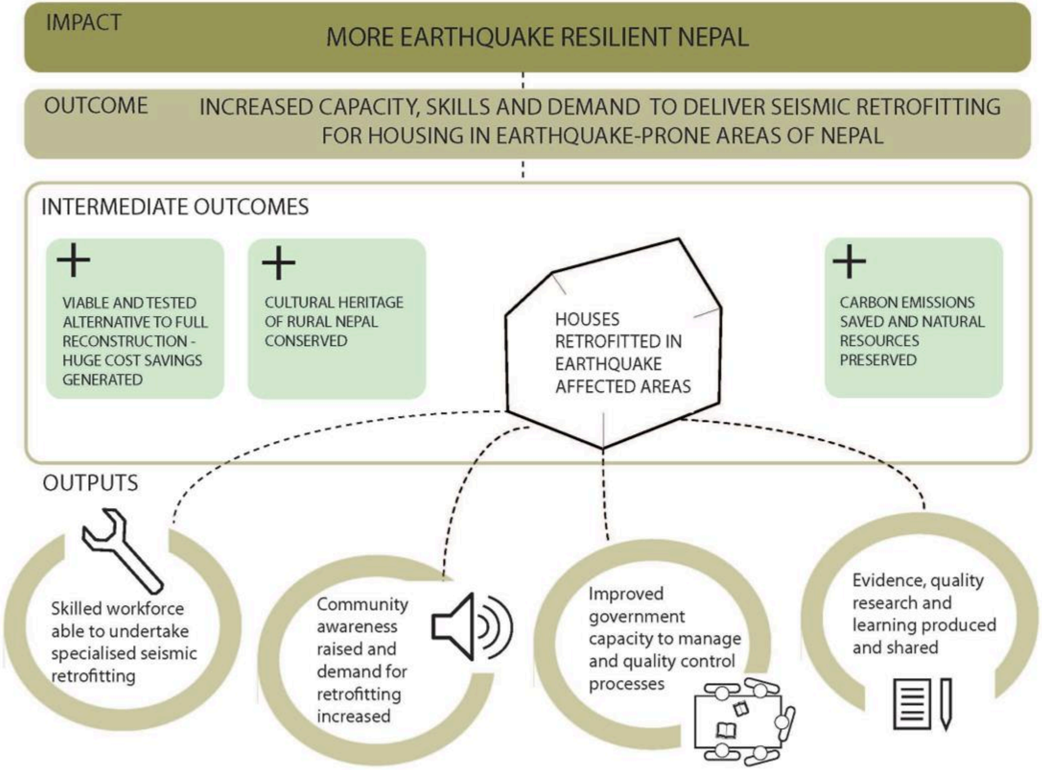
# Springboard Project – Training

CLPIU orientation session in Dhulikhel, Kavre, led by Lead Structural Engineer Liva Shrestha

# SEISMIC RETROFITTING OF UNSAFE HOUSING IN NEPAL

Department for International Development

UKaid from the British people




## SEISMIC RETROFITTING OF UNSAFE HOUSING IN NEPAL

## THEORY OF CHANGE

# Consortium

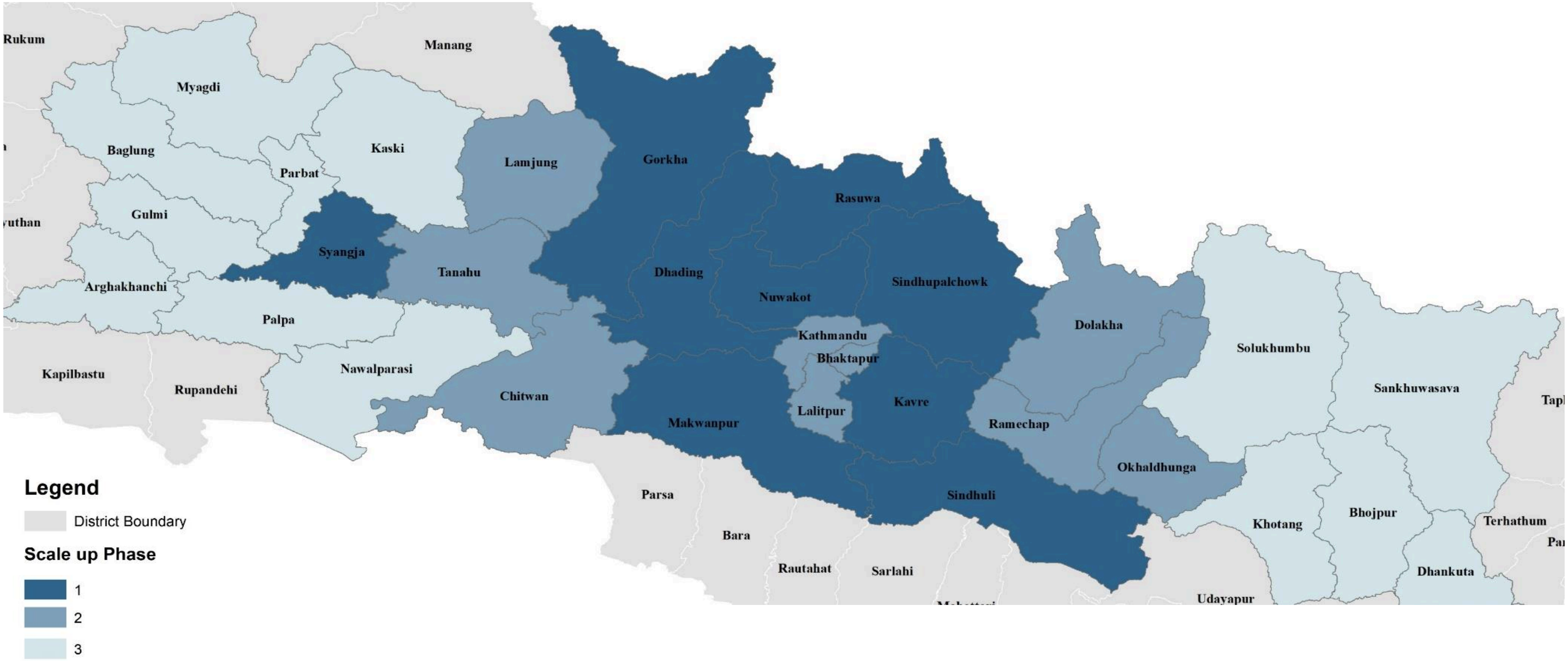
UNOPS

BBC MEDIA ACTION
TRANSFORMING LIVES THROUGH MEDIA AROUND THE WORLD

BUILD CHANGE

tearfund

# Preliminary Proposal for Scale Up Phases



# Outputs & Indicators

| Output | Indicator |
|---|---|
| 1. Strengthen Supply | Model/OJT retrofits<br>On-the-job training<br>Competency based training |
| 2. Generate Demand | Radio & television programming & PSAs<br>Community theatre & outreach |
| 3. Ensure Quality | GoN Inspectors - 7-day training<br>Theoretical & OJT training<br>6 month engineering internships |
| 4. Capture & Disseminate Learning | Grant fellowships<br>DRR platform<br>Effective communication for retrofitting report |

# Next Steps - January

- Official project launch
- Nationwide communication on retrofitting led by BBC Media Action
- Wider stakeholder communication on district-wise roll out plan
- Finalisation with CLPIU on next round of field inspector training

For technical information on retrofitting please contact

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